

Shared separations, shifted peaks, and upper-limit troughs

Standalone continuation of the v5.2 signal-structure study

Version 5.2.2 — 10 September 2026

What is tested

Can two positive components, separated by roughly 25 or 29 MeV and shifted slightly between campaigns, produce the troughs in the observed single-signal upper limits? This study fits those mass families and then propagates complete fitted and injected spectra through the moving-mask GP analysis. It varies separation, total size and component balance, and recomputes actual 90% CL_s limits every 0.5 MeV.

The lower family now specifically covers the features near 66 and 90 MeV. It is distinct from the earlier 65/80 MeV comparison. The upper family covers the 90 and 120 MeV features in 2016 and 2021. The native 2021 sample remains 10%; the 2015 hypothesis range remains extended to 100 MeV for this exploratory study.

Main result

Positive injected pairs can suppress the limits between their peaks in these datasets. The depth depends on spacing, yield balance and continuum choice. The moving GP training response accounts for part of the effect; a smaller response can remain when training is frozen because the correlated background nuisance is still profiled. A dip is therefore an estimator response to test, not an automatic consequence of two resolved particles.

Family	Search	min U/U_0 , GP updated	min U/U_0 , training frozen
Lower	2015	0.912	0.935
Lower	2016	0.691	0.888
Lower	2021 (10%)	0.750	0.941
Lower	All three	0.737	0.904
Upper	2016	0.663	0.875
Upper	2021 (10%)	0.705	0.947
Upper	2016 + 2021	0.618	0.905

The table uses the fitted shared-rate pair at unit strength, and the central third of its between-peak interval. Each control's minimum is taken within that same fixed interval; the two minima need not occur at the same mass. U_0 is the response of the fitted continuum alone. Ratios below one indicate additional tightening relative to that baseline, not the significance of a signal.

The fitted shared separations are 24.75 MeV (lower all three), 24.60 MeV (lower 2016+2021) and 29.19 MeV (upper 2016+2021). Lower-family mass shifts improve the likelihood very little, while shared-rate and continuum assumptions materially affect the reconstruction. No new toys, global significance or coverage calibration are introduced.

1. Fixed experiments and shifted mass hypotheses

Within each family and campaign, all mass/rate hypotheses use identical observed bins, external-GP mean and covariance. The lower reference midpoint is restricted to 76–81 MeV and separation to 20–30 MeV; the upper ranges are 100–106 and 22–32 MeV. The fixed omitted region includes every allowed pair plus three resolutions. Kernel anchors are 78.5 and 103 MeV. Relative campaign offsets satisfy $|\delta_d| \leq 3$ MeV, with the first campaign fixed at zero.

$$m_{d,1} = C + \delta_d - \Delta/2, \quad m_{d,2} = C + \delta_d + \Delta/2.$$

The strict model fixes every offset to zero; the shifted model shares Δ ; the free-pair model permits separate campaign midpoints and separations. The free model has an expanded absolute midpoint domain for nonreference years, rather than retaining the shifted model’s relative-offset bound. Fits enumerate nonnegative-amplitude faces. Multiple deterministic starts and local probes check the selected solution; they do not prove a unique global optimum over all possible signal models.

Family	Fit	Campaign	m_1 [MeV]	m_2 [MeV]	δ_d [MeV]
Lower	All three	2015	64.332	89.085	0.000
Lower	All three	2016	67.332	92.085	3.000
Lower	All three	2021	66.846	91.599	2.514
Lower	16+21	2016	67.421	92.025	0.000
Lower	16+21	2021	66.912	91.515	-0.509
Upper	16+21	2016	90.005	119.198	0.000
Upper	16+21	2021	92.824	122.016	2.818

Case	Allow shifts	Release spacing	Release rates
Lower all three	0.568	1.331	19.981
Lower 16+21	0.088	0.995	4.367
Upper 16+21	3.317	0.787	9.097

Entries are $2\Delta \log L$ on the same fixed experiment, not p-values. Rate comparisons allow the mass coordinates to reoptimize in each rate model. The lower all-three shifted fit reaches the +3 MeV 2016 offset bound. The upper strict fit reaches its 106 MeV midpoint bound. The small lower shift gains do not establish a calibration discrepancy; shared reconstructed spacing does not establish compatible rates or a common particle origin.

2. Injection normalization and the source of a dip

The main continuum B_d is the positive full-support extension of the shared-rate shifted fit. Let $g_d(m_j)$ be a bin-integrated Gaussian normalized over the complete stored histogram. The deterministic mean is

$$\bar{n}_d = B_d + a \sum_{j=1}^2 N_{dj}^{\text{ref}} g_d(C + \delta_d \pm \Delta/2).$$

At the fitted separation and $a = 1$, this reproduces the unmodified fitted mean. No amplitude floor is used. For separation scans, each full-support event yield N_{dj}^{ref} is held fixed: a mass-dependent conversion cannot masquerade as a separation effect. The implied coupling then changes with mass, so these injections are morphology experiments. Fixed-coupling controls retain the fitted common amplitudes instead. The extraction always uses the retained shared-coupling template for the tested single signal.

Reference fit	Campaign	N_1^{ref} [events]	N_2^{ref} [events]
Lower all three	2015	724.2	62.3
Lower all three	2016	10449.0	5802.2
Lower all three	2021	13110.8	15070.9
Lower 16+21	2016	9939.0	5622.9
Lower 16+21	2021	12503.8	14579.1
Upper 16+21	2016	8314.3	1442.9
Upper 16+21	2021	20277.9	8222.0

Spacing values are 12, 18, 24, 30 and 36 MeV plus the fitted value; strength multipliers are 0, 0.5, 1 and 2. Additional controls use high/low yield ratios 0.5, 1 and 2 at fixed total yield, or retain only one original component. These balance choices are distinct from multiplying the original fitted ratio.

At each tested mass, the main analysis recomputes GP predictions from the injected sidebands. Peaks outside the moving exclusion can raise the interpolation under its center. If that increase exceeds the injected in-window response along the tested template, the single-signal fit becomes more negative and its upper limit can tighten. The paired frozen-training control holds the baseline GP mean and covariance fixed, while keeping exactly the same injected fit-bin counts and tested template. Background nuisances are still profiled; this control does not guarantee a perfectly flat or monotonic response.

The external-GP control removes the fitted continuum displacement. The independent-rate reconstruction changes continuum, component strengths and optimized positions together; it is an alternative fitted spectrum, not an isolated rate intervention. A zero upper component remains zero in its 2021 upper-family fit.

3. Aligned likelihoods, upper limits and diagnostic scores

The horizontal reference coordinate is u . In campaign d , the tested mass is $u + \delta_d$; its moving $\pm 2.25\sigma_d$ exclusion, signal template and kernel state are rebuilt there before combining likelihoods. Campaign counts and backgrounds are stacked with independent GP nuisance blocks and one shared amplitude $A = \epsilon^2/10^{-8}$:

$$\lambda = b + L\theta + AS, \quad \ell = \sum_i [\lambda_i - n_i + n_i \log(n_i/\lambda_i)] + \frac{1}{2}\theta^T \theta.$$

The mean must be positive in occupied bins. The free-amplitude fit can be negative; nonnegative tested amplitudes use the physical bounded likelihood-ratio denominator. The upper limit solves $\text{CL}_s = 0.1$ with separately profiled observed and background-Asimov likelihood ratios. Writing those ratios as q and q_a , the inherited bounded asymptotic tails [2] are

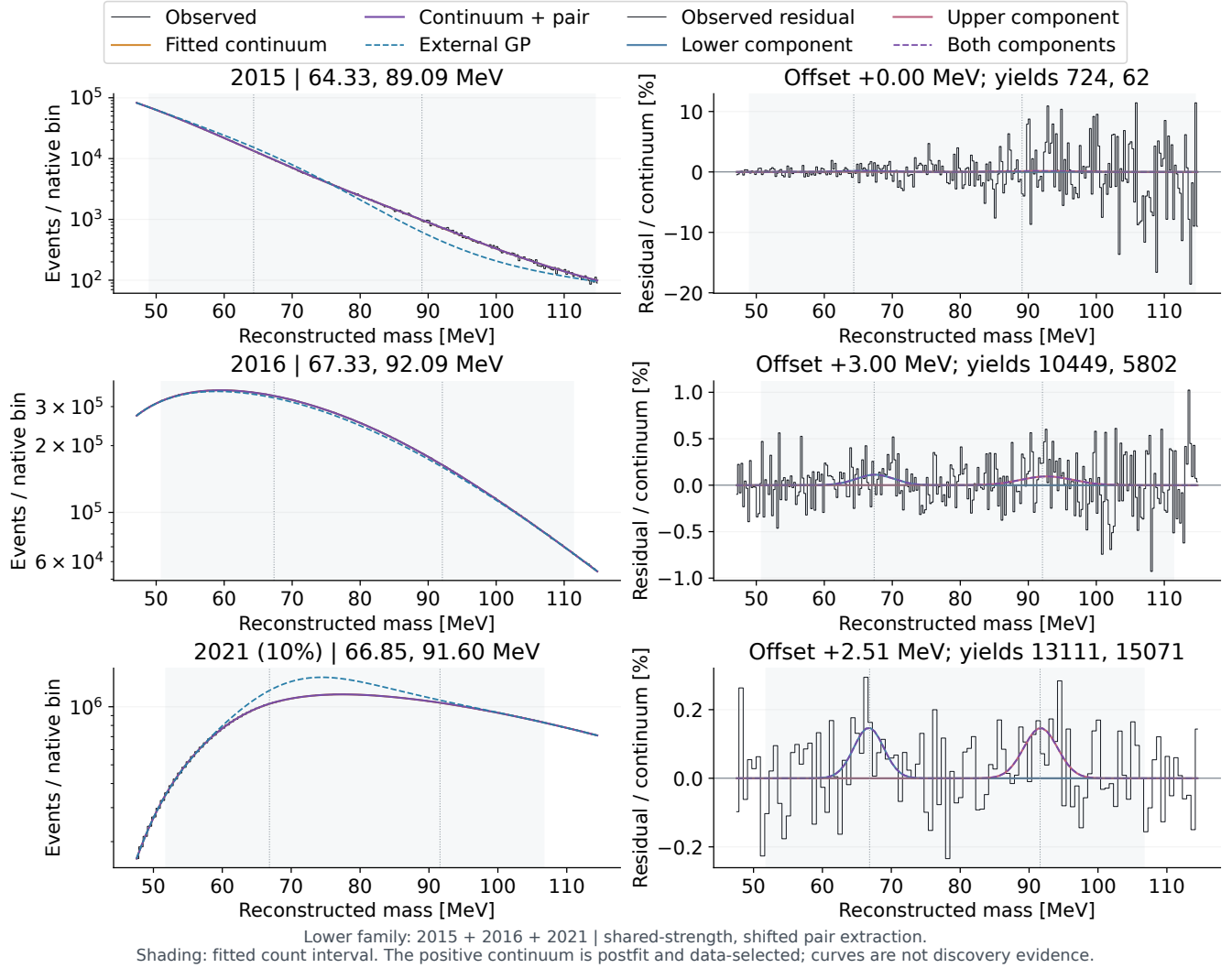
$$(z_{s+b}, z_b) = \begin{cases} (\sqrt{q}, \sqrt{q} - \sqrt{q_a}), & q \leq q_a, \\ ((q + q_a)/(2\sqrt{q_a}), (q - q_a)/(2\sqrt{q_a})), & q > q_a, \end{cases} \quad \text{CL}_s = \Phi(-z_{s+b})/\Phi(-z_b).$$

The signed root $r = \text{sign}(\hat{A})\sqrt{2(\ell_0 - \ell_{\text{free}})}$ retains deficits. Its positive-signal reference is $p_0 = \Phi[-\max(0, r)]$, with deficits at 0.5. Applying this transform to a deterministic mean spectrum is a score diagnostic, not a measured probability. Neither fixed-mass references nor the older stress-centered significance GP calibrate the selected shifted-pair search. The prior field can motivate regions, but its covariance and trials factors are not transferred here.

Kernel parameters are fixed at archived states and interpolated in their logarithms; GP predictions and count-dependent noise are updated, while hyperparameters are not reoptimized. Native bins, resolution and observed-density conversion remain unchanged. The lower reference window is 50–112 MeV and upper window 75–140 MeV, sampled every 0.5 MeV. Any coordinate outside a constituent’s mass domain is omitted without changing that combination’s membership. The 2015 domain ends at 100 MeV and its kernel is clamped to the archived 90 MeV state above 90. This is not a newly qualified extension.

Dip metrics use the predeclared central third $[C - \Delta/6, C + \Delta/6]$. The ratio is U/U_0 at matched coordinates, using each external-pair control’s own external baseline. The geometric flag $\Delta > 2.25(\sigma_1 + \sigma_2)$ means the two specified exclusion windows do not overlap; it is not a demonstration that the detector resolves two resonances. Whole-window and outside-peak log-limit RMS differences quantify descriptive shape agreement with data, with no independence or chi-square interpretation.

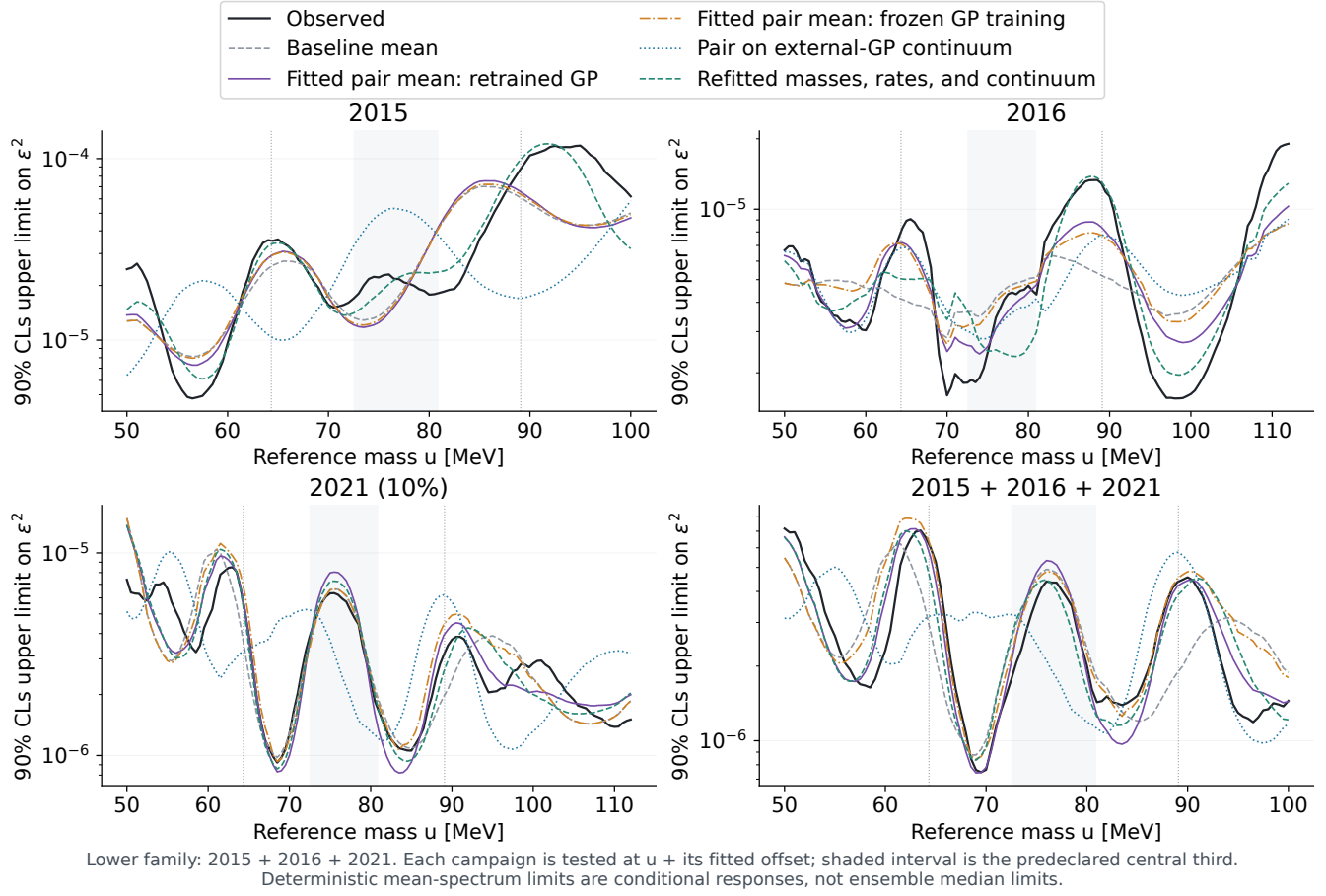
4. 66/90 MeV family: all three campaigns – extracted components



Observed bins and full fitted spectra in reconstructed mass. The continuum and each positive component are displayed separately; background displacement from the external GP is visible. Full-support tails are retained in every propagation. Counts are analysis-bin counts, not a new event selection.

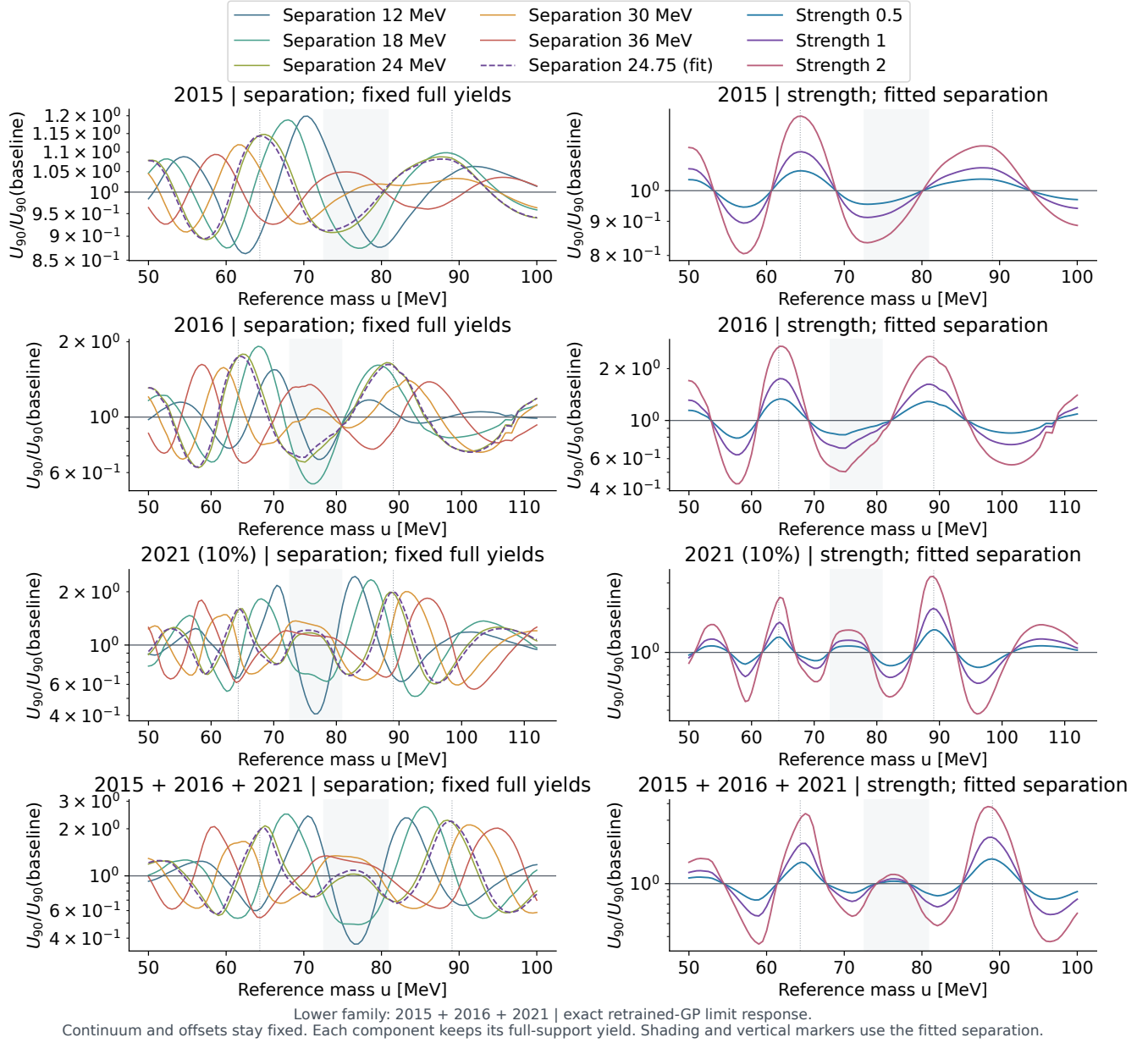
The lower all-three continuum differs from its external-GP mean by up to +68.24% in 2015 and −19.40% in 2021 over the stored support. The upper-family shared continuum differs by about −5.15% in 2016 and −2.65% in 2021 at its largest signed displacements. These selected baselines are part of the explanation being tested, not independent background measurements.

5. 66/90 MeV family: all three campaigns – absolute upper limits



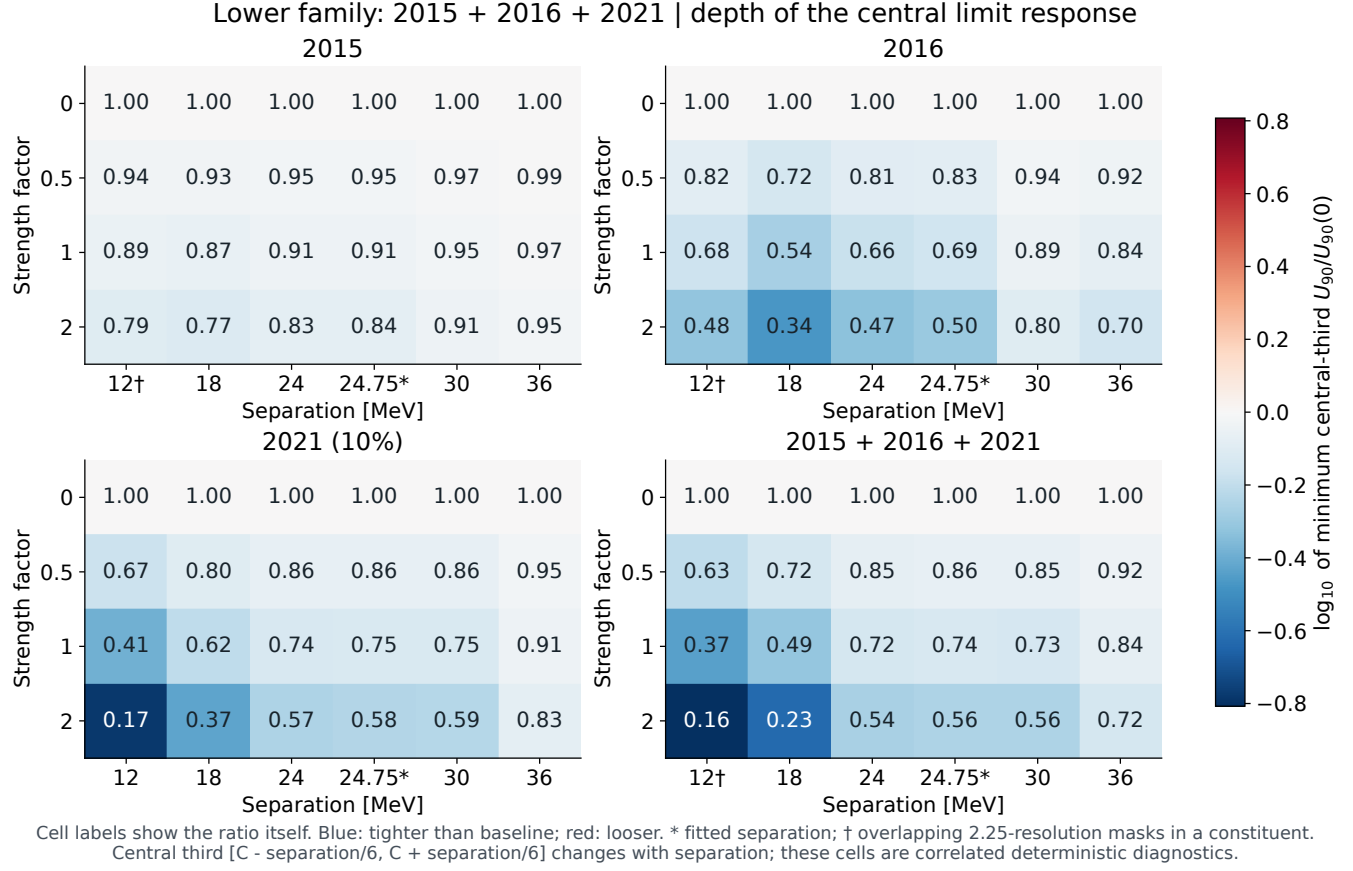
Actual 0.5 MeV CL_s solutions from observed data, the continuum alone, the fitted shared-rate mean with GP training updated or frozen, the external-GP pair, and the independently refitted spectrum. The coordinate u is mapped to each campaign as specified in the mass table. Deterministic curves are full-pipeline responses, not independently validated ensemble medians.

6. 66/90 MeV family: all three campaigns – spacing and size



Ratios to the matched fitted-continuum baseline. Separation varies at unit strength while reference event yields and midpoint are fixed; strength varies at the fitted separation. Vertical peak guides refer to the selected pair unless a panel explicitly shows a different spacing. A deeper bridge response can coexist with a poorer description elsewhere in the mass window.

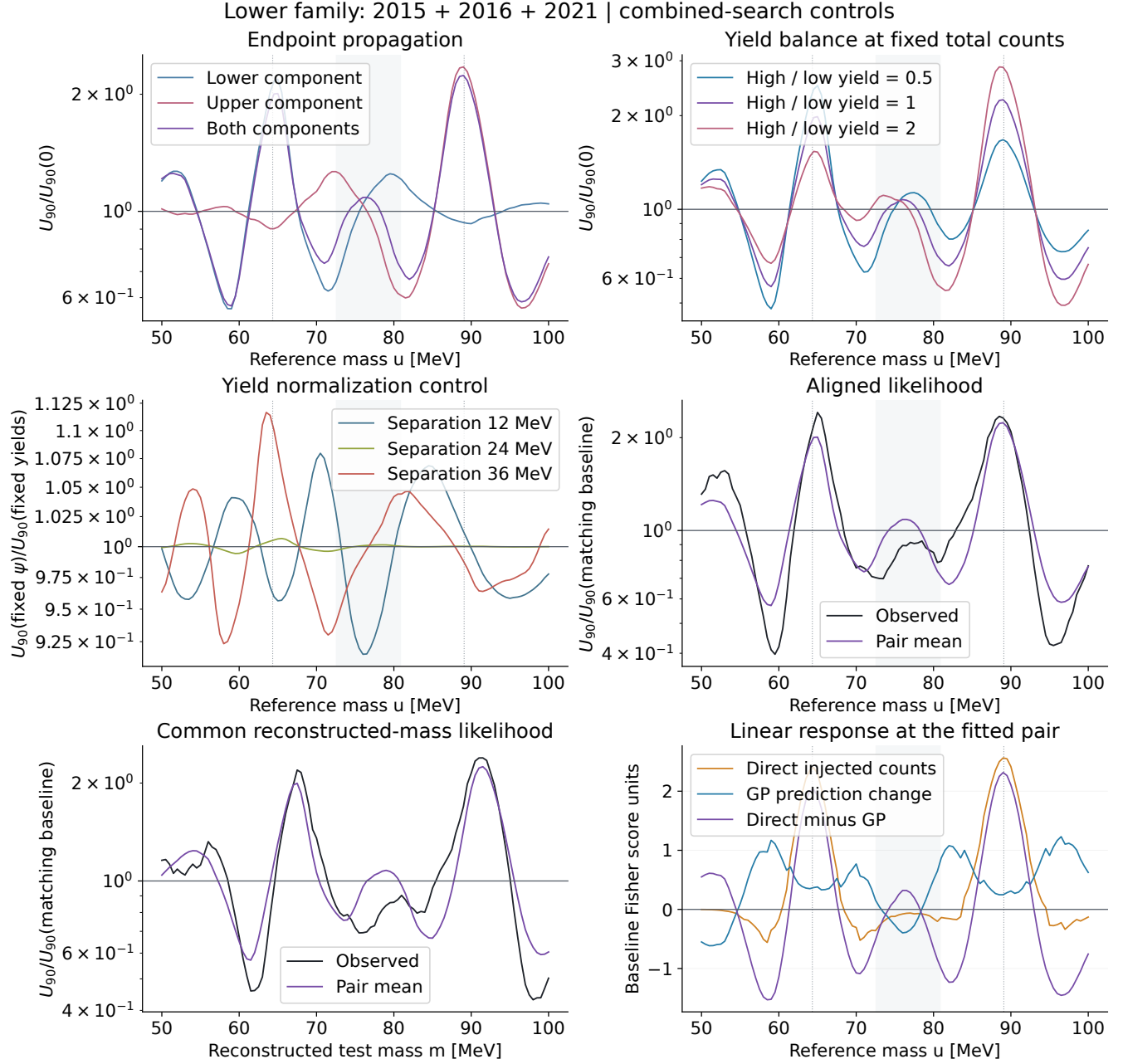
7. 66/90 MeV family: all three campaigns – dip catalogue



Minimum limit ratio in the central third of the interval between the injected peaks, as a function of separation and strength. The zero-strength row/column is the common baseline. Every nonzero cell is obtained by fitting the full injected histogram at each tested mass; interpolation is only graphical. This catalogue ranks dip response, not signal likelihood or statistical significance.

Within this declared grid, the largest combined tightening is $U/U_0 = 0.156$ at separation 12.00 MeV and strength 2.0. This is a response extremum among the simulated means, not a fitted estimate of separation or signal size.

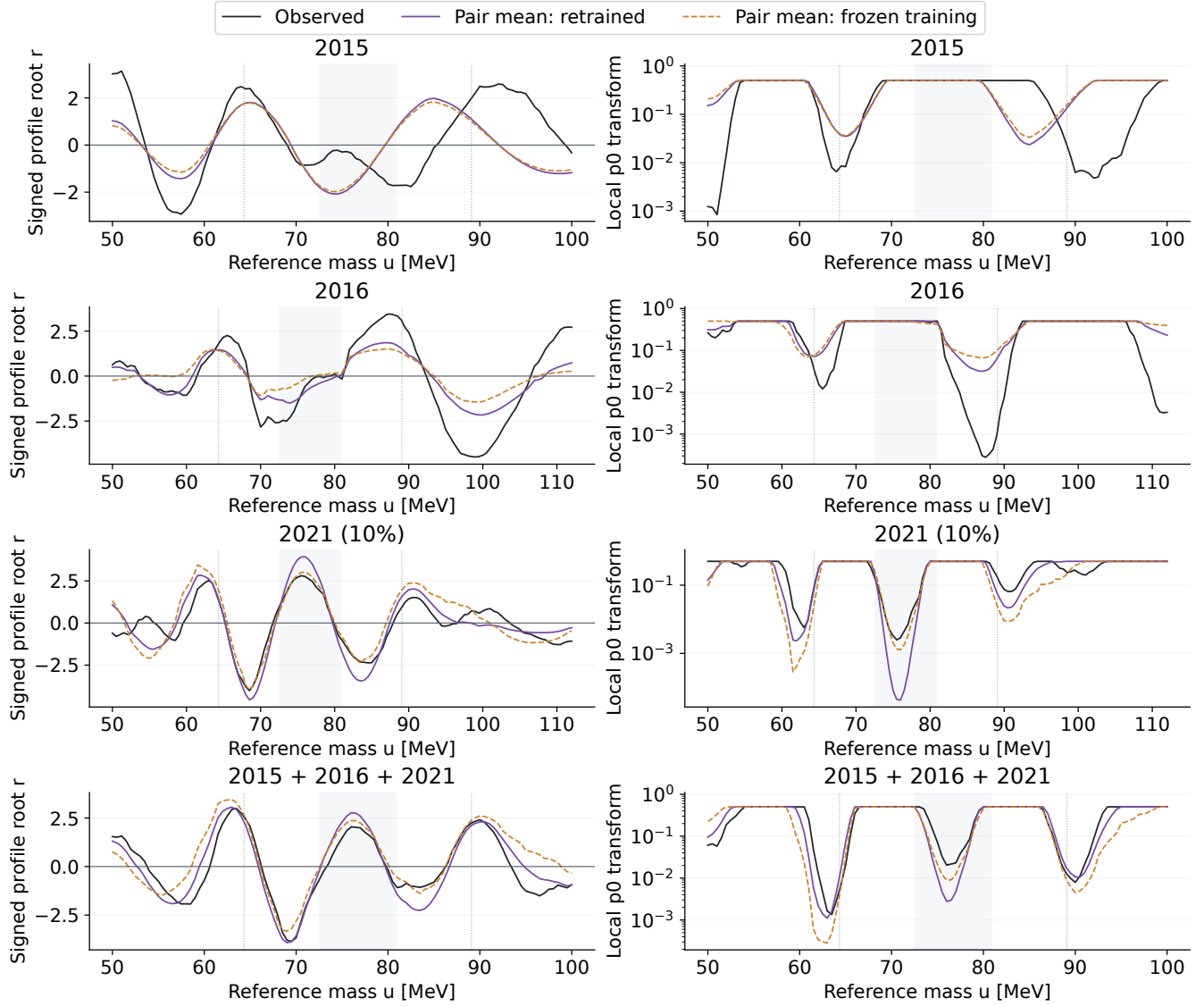
8. 66/90 MeV family: all three campaigns – mechanism and controls



The bottom-left panel uses common reconstructed mass m . All other panels use the aligned reference coordinate u . Each ratio uses its matching baseline. The linear score projection explains the mechanism; upper limits above use the exact nonlinear solver.

Component, balance, normalization and alignment controls. Fixed event yield isolates geometry; fixed coupling permits the conversion to change the generated event yield. Raw and aligned scans use their explicitly labeled coordinates, and aligned combinations are recomputed likelihoods. Any linear score decomposition is a baseline-covariance diagnostic, while the upper-limit curves use the exact profiled count objective.

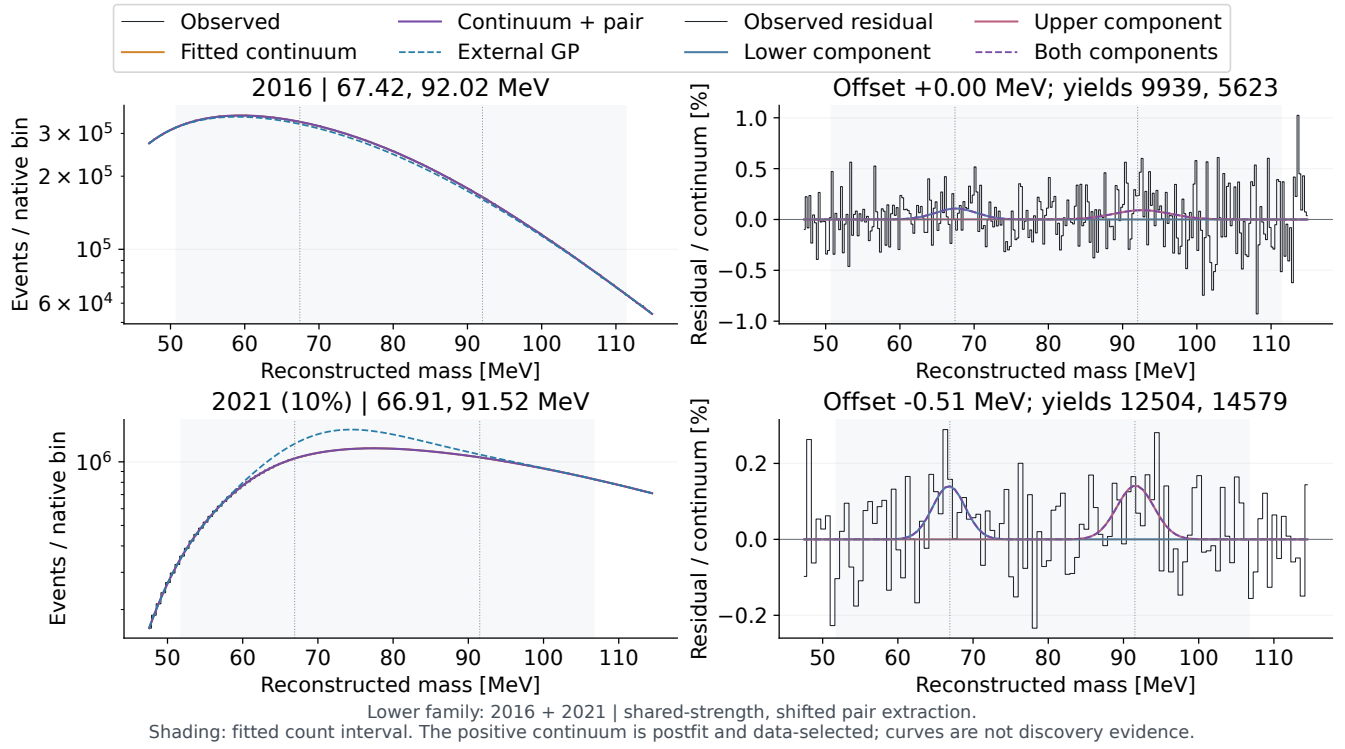
9. 66/90 MeV family: all three campaigns – signed roots and local references



Lower family: 2015 + 2016 + 2021. Observed p_0 is a conditional fixed-mass diagnostic; fitted-mean p_0 describes response only. Deficits have $p_0 = 0.5$. No curve calibrates the selected pair, separation, offsets, or a global discovery probability.

Signed-root curves expose the negative response hidden by positive-signal p_0 . Data use the fixed-mass asymptotic reference; mean-spectrum transforms are diagnostic only. Adjacent masses reuse counts and GP sidebands, so a peak, bridge dip and second peak cannot be assigned independent probabilities and multiplied. No updated calibration of the significance GP or shifted search is claimed.

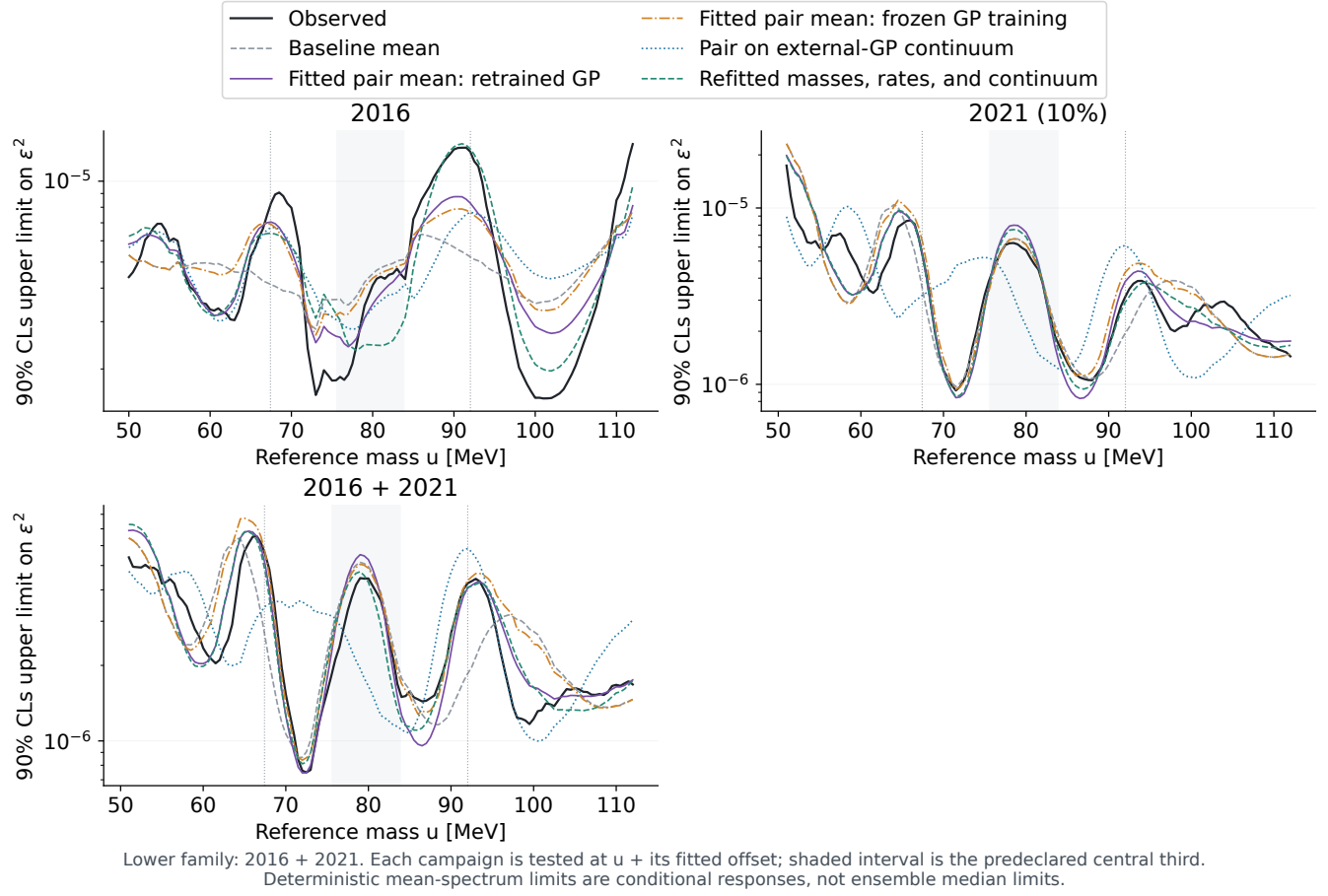
10. 66/90 MeV family: 2016 + 2021 – extracted components



Observed bins and full fitted spectra in reconstructed mass. The continuum and each positive component are displayed separately; background displacement from the external GP is visible. Full-support tails are retained in every propagation. Counts are analysis-bin counts, not a new event selection.

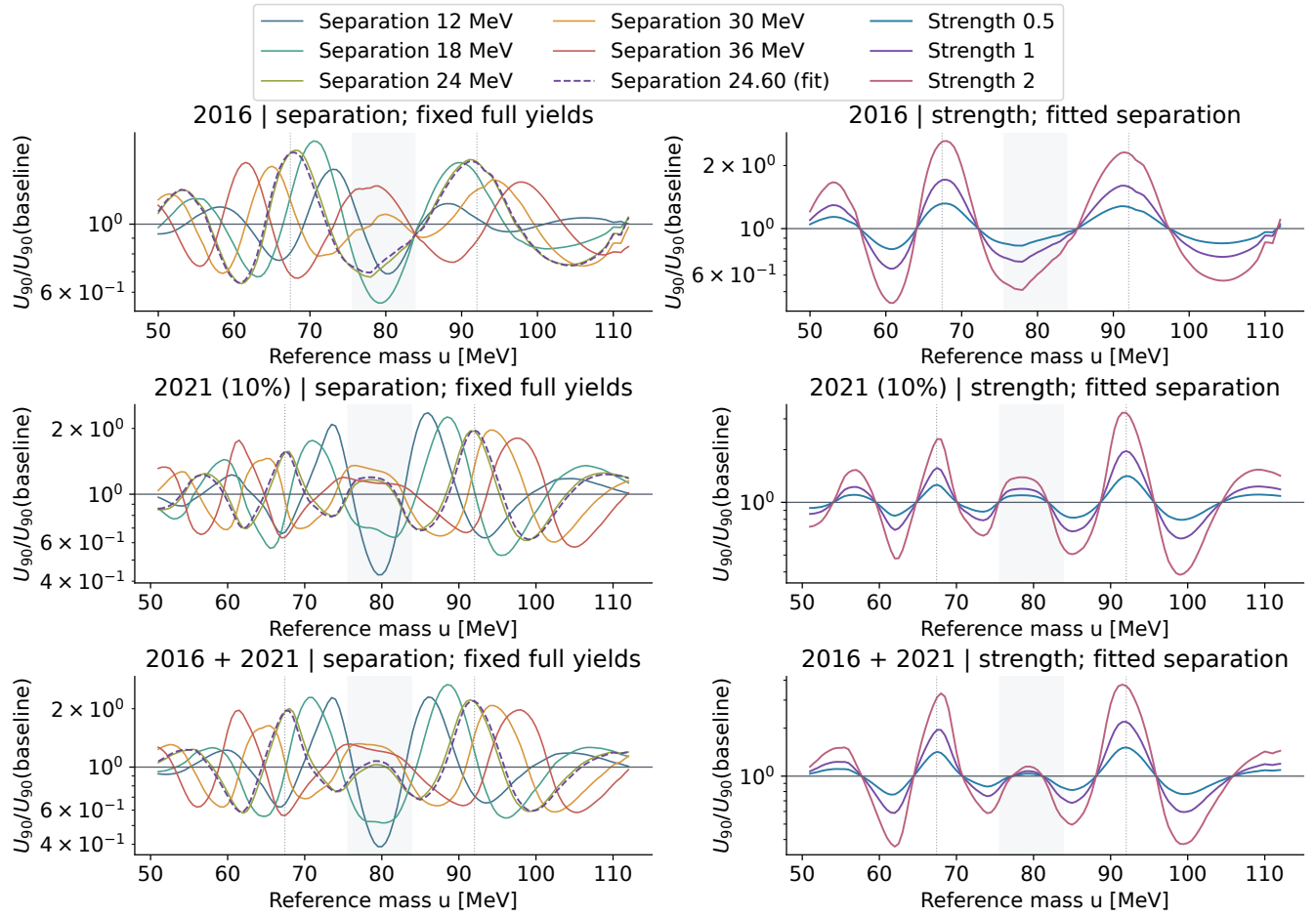
The same positive continuum is retained across all separation and strength variations in this case. The reference yields come from the shared-rate fit; alternative independent-rate reconstructions retain any zero fitted component without replacing it by a benchmark floor.

11. 66/90 MeV family: 2016 + 2021 – absolute upper limits



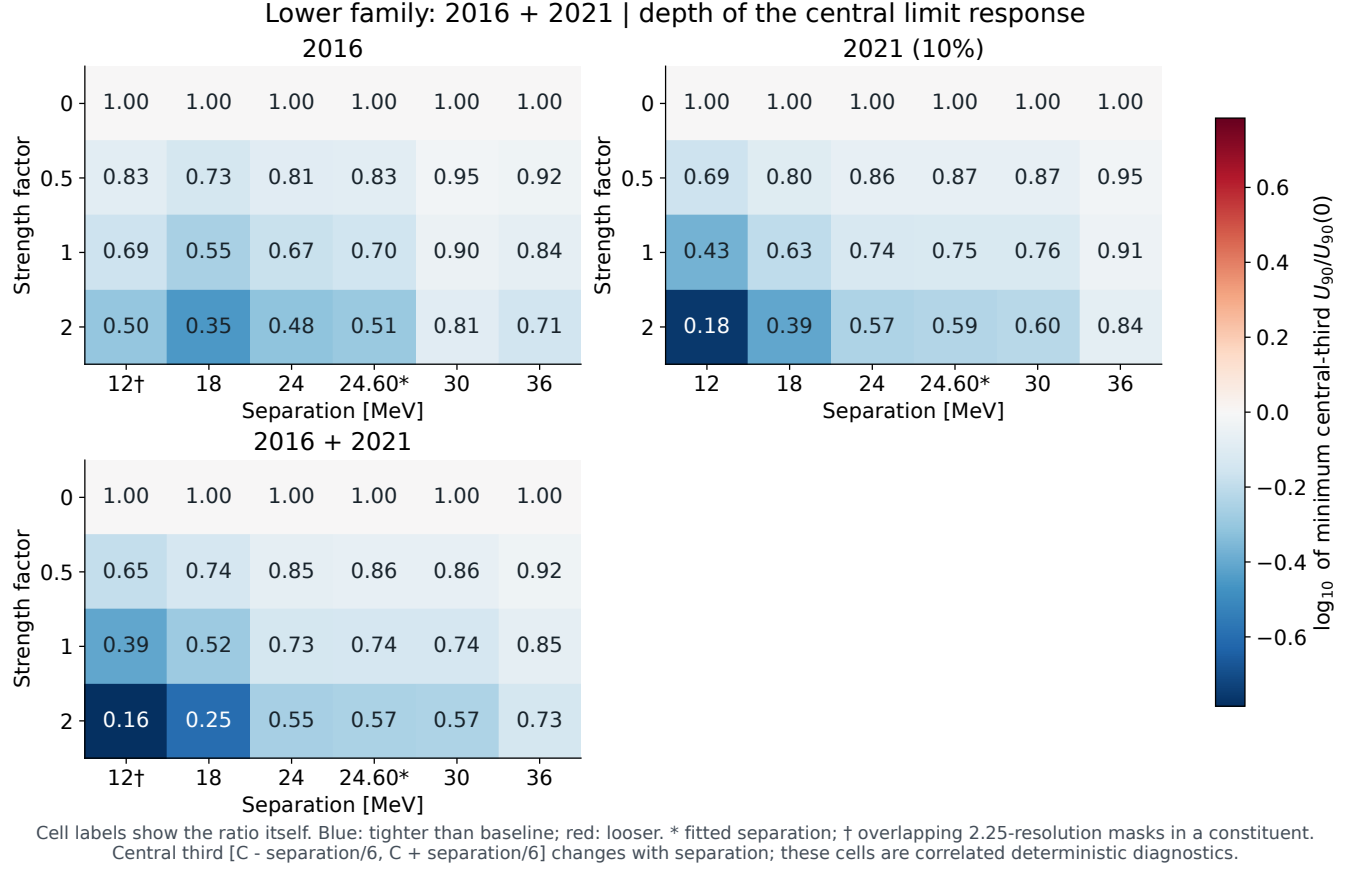
Actual 0.5 MeV CL_s solutions from observed data, the continuum alone, the fitted shared-rate mean with GP training updated or frozen, the external-GP pair, and the independently refitted spectrum. The coordinate u is mapped to each campaign as specified in the mass table. Deterministic curves are full-pipeline responses, not independently validated ensemble medians.

12. 66/90 MeV family: 2016 + 2021 – spacing and size



Ratios to the matched fitted-continuum baseline. Separation varies at unit strength while reference event yields and midpoint are fixed; strength varies at the fitted separation. Vertical peak guides refer to the selected pair unless a panel explicitly shows a different spacing. A deeper bridge response can coexist with a poorer description elsewhere in the mass window.

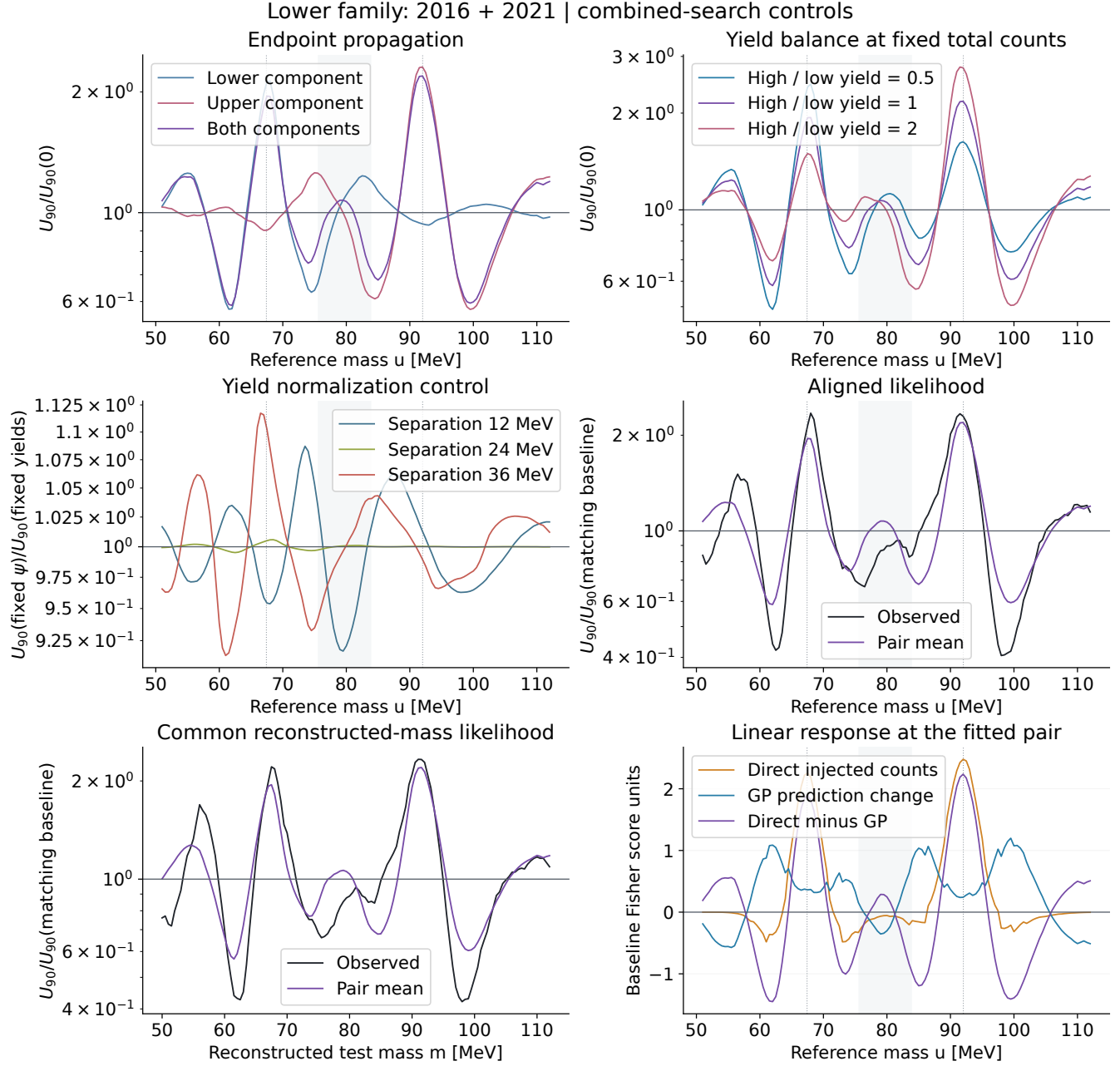
13. 66/90 MeV family: 2016 + 2021 – dip catalogue



Minimum limit ratio in the central third of the interval between the injected peaks, as a function of separation and strength. The zero-strength row/column is the common baseline. Every nonzero cell is obtained by fitting the full injected histogram at each tested mass; interpolation is only graphical. This catalogue ranks dip response, not signal likelihood or statistical significance.

Within this declared grid, the largest combined tightening is $U/U_0 = 0.164$ at separation 12.00 MeV and strength 2.0. This is a response extremum among the simulated means, not a fitted estimate of separation or signal size.

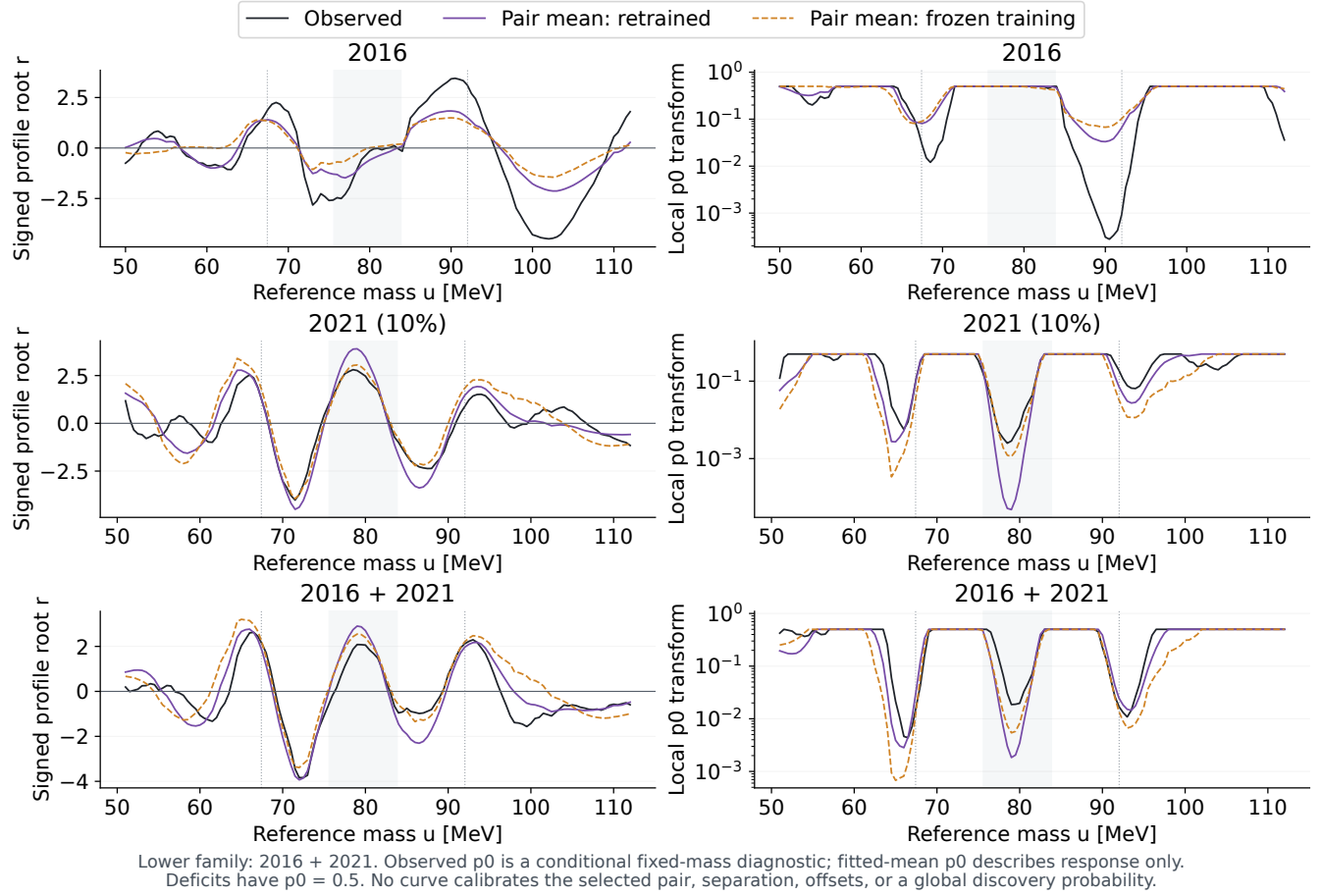
14. 66/90 MeV family: 2016 + 2021 – mechanism and controls



The bottom-left panel uses common reconstructed mass m . All other panels use the aligned reference coordinate u . Each ratio uses its matching baseline. The linear score projection explains the mechanism; upper limits above use the exact nonlinear solver.

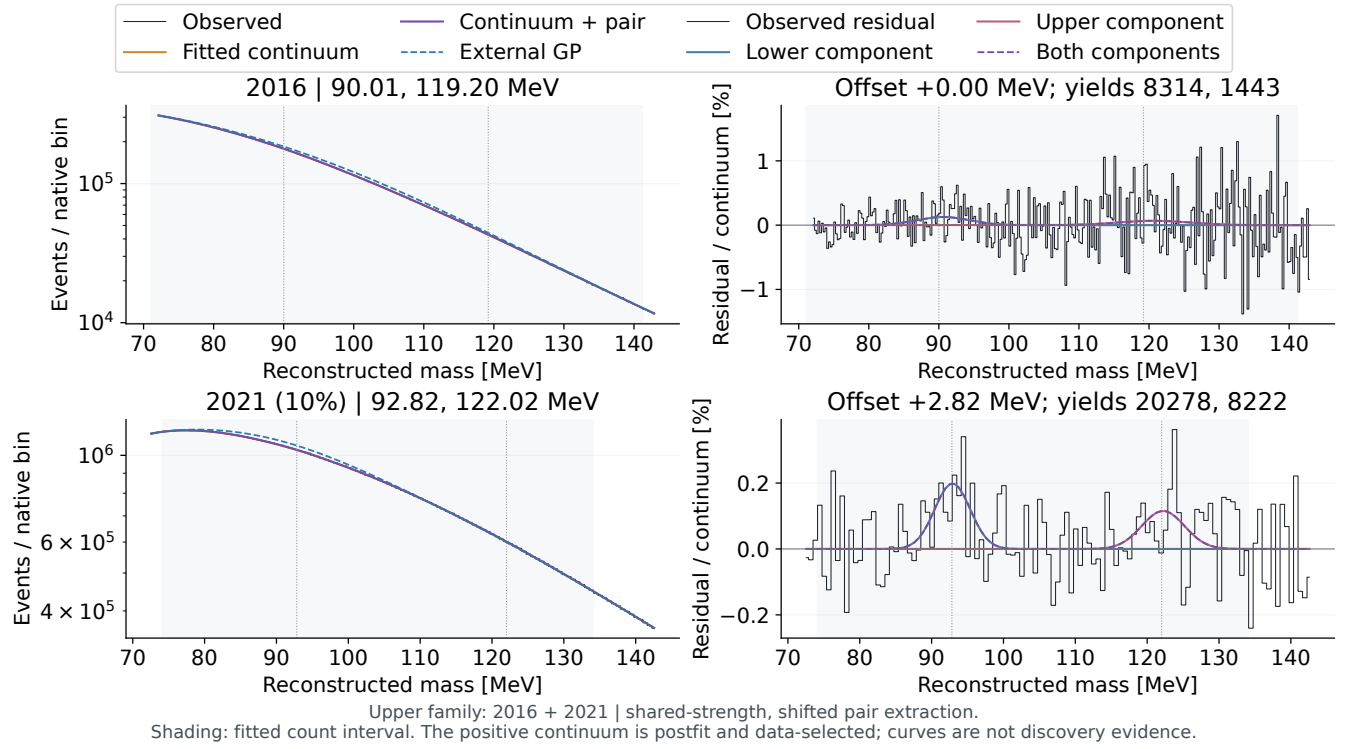
Component, balance, normalization and alignment controls. Fixed event yield isolates geometry; fixed coupling permits the conversion to change the generated event yield. Raw and aligned scans use their explicitly labeled coordinates, and aligned combinations are recomputed likelihoods. Any linear score decomposition is a baseline-covariance diagnostic, while the upper-limit curves use the exact profiled count objective.

15. 66/90 MeV family: 2016 + 2021 – signed roots and local references



Signed-root curves expose the negative response hidden by positive-signal p_0 . Data use the fixed-mass asymptotic reference; mean-spectrum transforms are diagnostic only. Adjacent masses reuse counts and GP sidebands, so a peak, bridge dip and second peak cannot be assigned independent probabilities and multiplied. No updated calibration of the significance GP or shifted search is claimed.

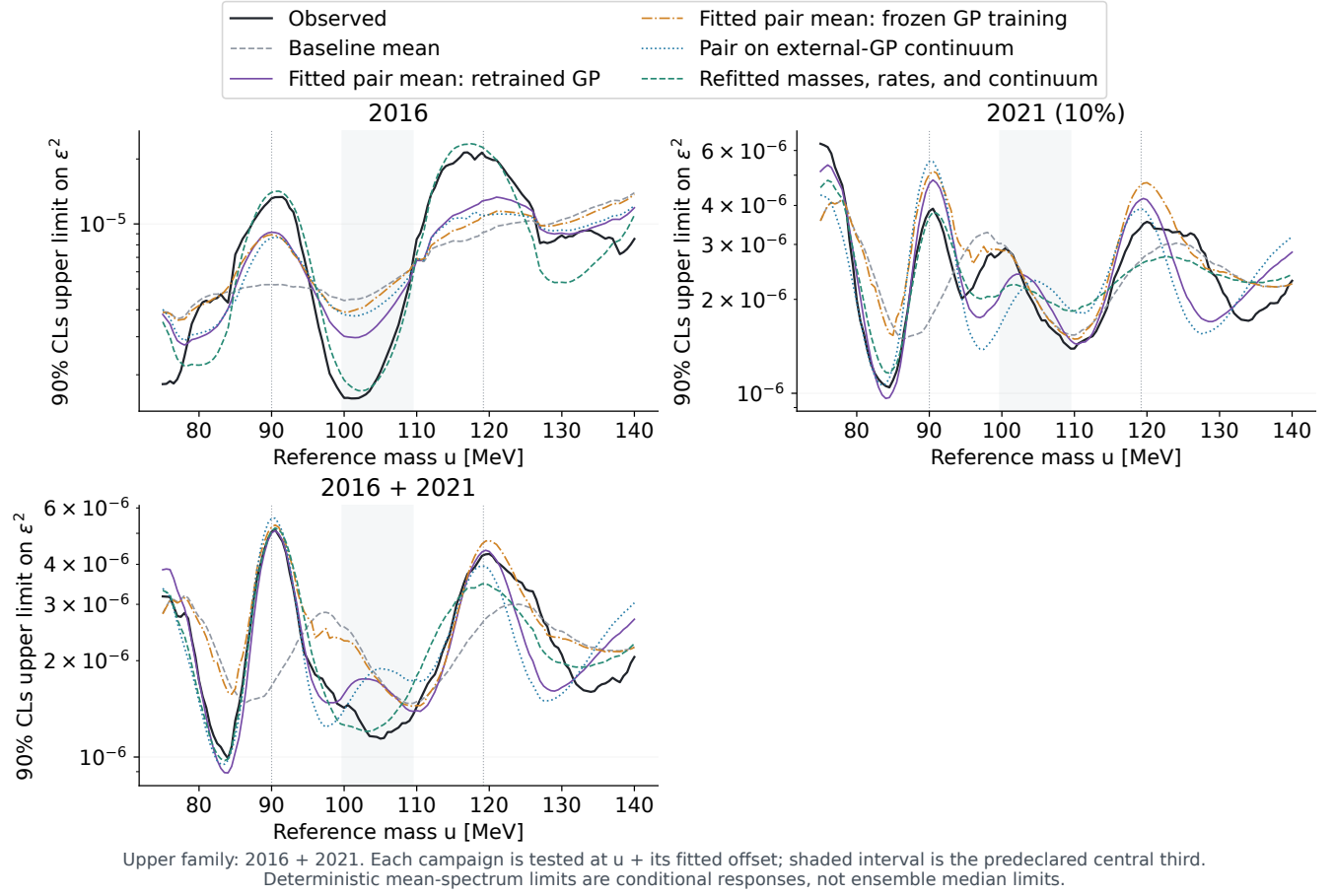
16. 90/120 MeV family: 2016 + 2021 – extracted components



Observed bins and full fitted spectra in reconstructed mass. The continuum and each positive component are displayed separately; background displacement from the external GP is visible. Full-support tails are retained in every propagation. Counts are analysis-bin counts, not a new event selection.

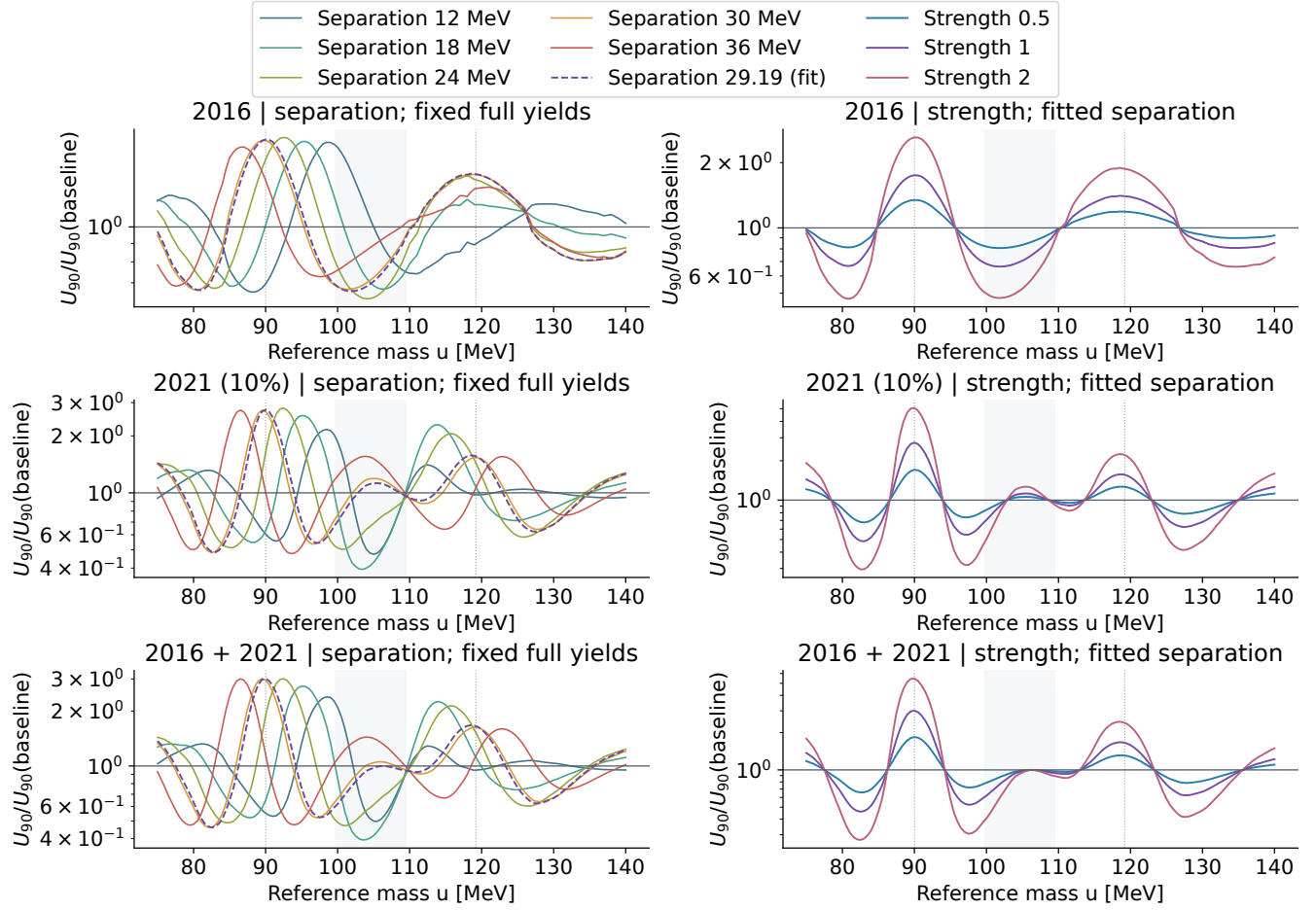
The same positive continuum is retained across all separation and strength variations in this case. The reference yields come from the shared-rate fit; alternative independent-rate reconstructions retain any zero fitted component without replacing it by a benchmark floor.

17. 90/120 MeV family: 2016 + 2021 – absolute upper limits



Actual 0.5 MeV CL_s solutions from observed data, the continuum alone, the fitted shared-rate mean with GP training updated or frozen, the external-GP pair, and the independently refitted spectrum. The coordinate u is mapped to each campaign as specified in the mass table. Deterministic curves are full-pipeline responses, not independently validated ensemble medians.

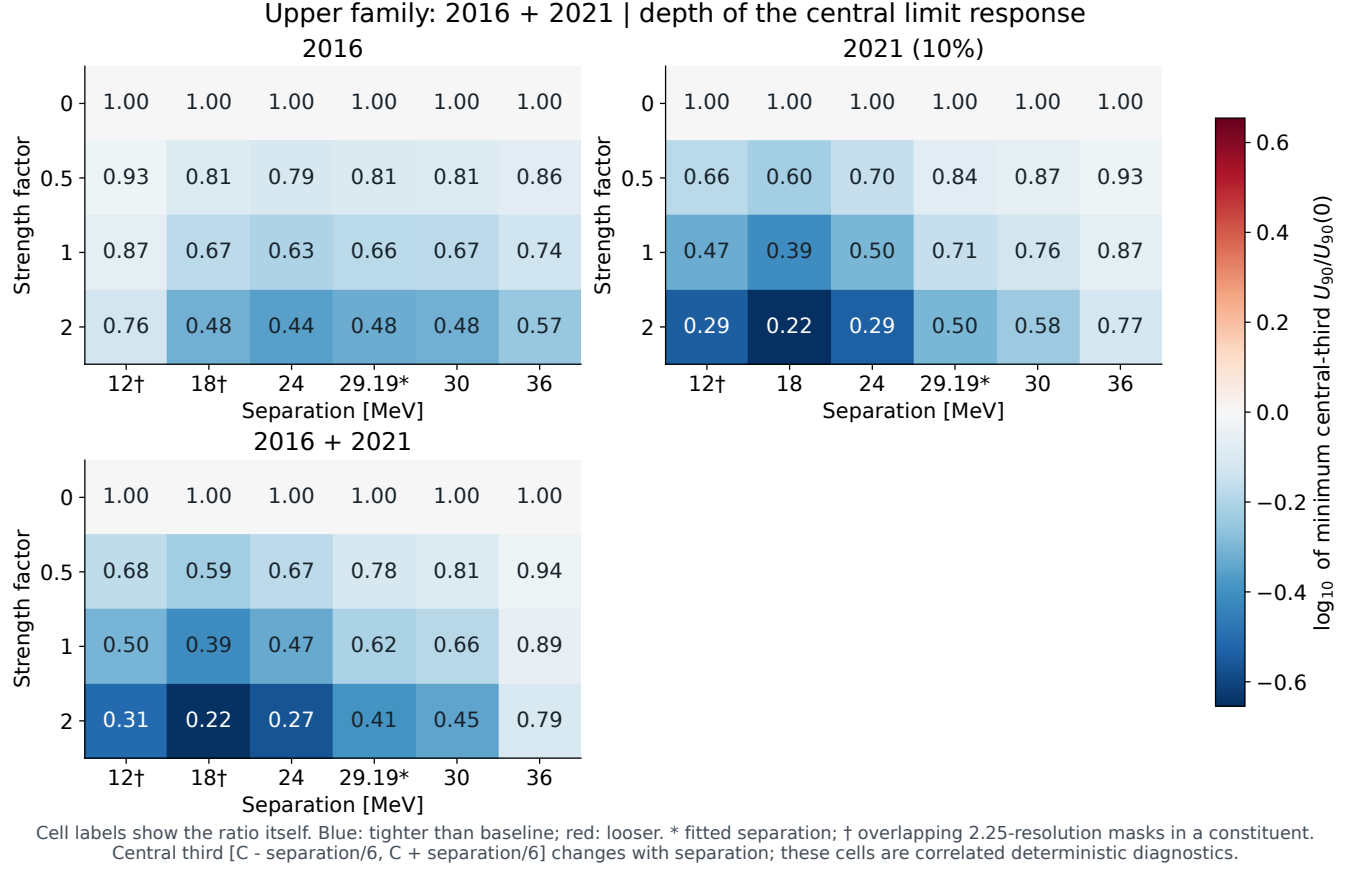
18. 90/120 MeV family: 2016 + 2021 – spacing and size



Upper family: 2016 + 2021 | exact retrained-GP limit response.
Continuum and offsets stay fixed. Each component keeps its full-support yield. Shading and vertical markers use the fitted separation.

Ratios to the matched fitted-continuum baseline. Separation varies at unit strength while reference event yields and midpoint are fixed; strength varies at the fitted separation. Vertical peak guides refer to the selected pair unless a panel explicitly shows a different spacing. A deeper bridge response can coexist with a poorer description elsewhere in the mass window.

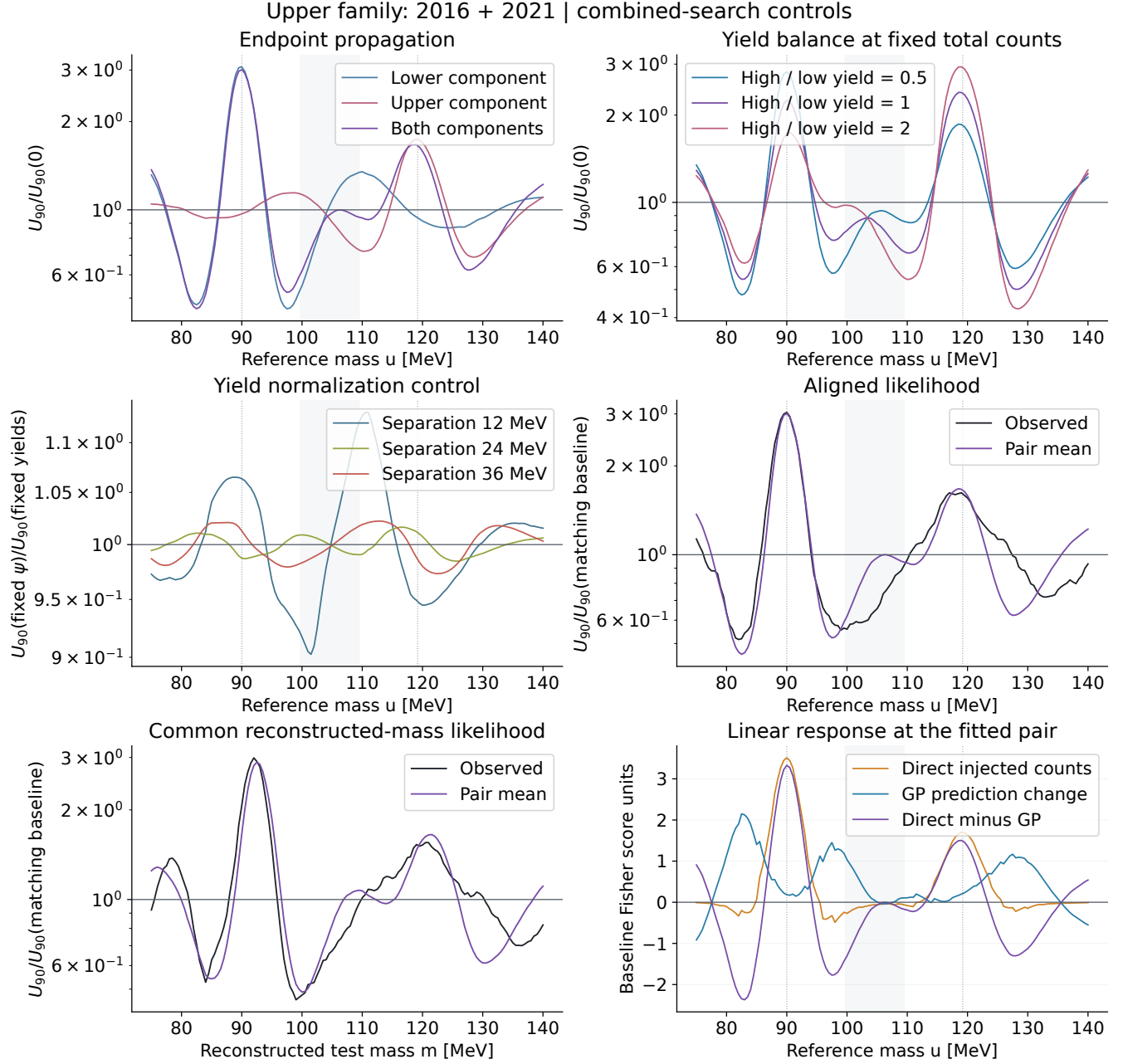
19. 90/120 MeV family: 2016 + 2021 – dip catalogue



Minimum limit ratio in the central third of the interval between the injected peaks, as a function of separation and strength. The zero-strength row/column is the common baseline. Every nonzero cell is obtained by fitting the full injected histogram at each tested mass; interpolation is only graphical. This catalogue ranks dip response, not signal likelihood or statistical significance.

Within this declared grid, the largest combined tightening is $U/U_0 = 0.225$ at separation 18.00 MeV and strength 2.0. This is a response extremum among the simulated means, not a fitted estimate of separation or signal size.

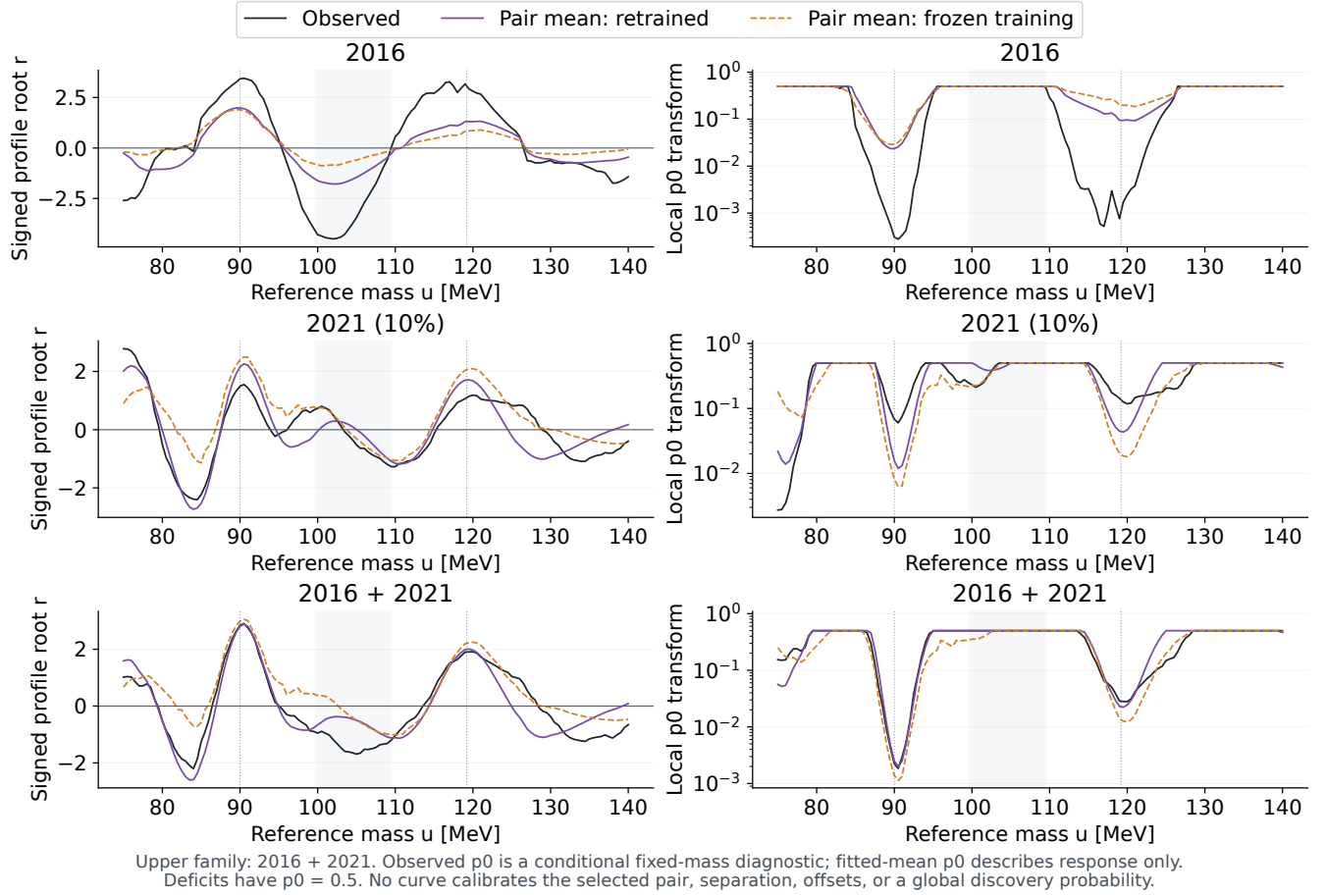
20. 90/120 MeV family: 2016 + 2021 – mechanism and controls



The bottom-left panel uses common reconstructed mass m . All other panels use the aligned reference coordinate u . Each ratio uses its matching baseline. The linear score projection explains the mechanism; upper limits above use the exact profiled count objective.

Component, balance, normalization and alignment controls. Fixed event yield isolates geometry; fixed coupling permits the conversion to change the generated event yield. Raw and aligned scans use their explicitly labeled coordinates, and aligned combinations are recomputed likelihoods. Any linear score decomposition is a baseline-covariance diagnostic, while the upper-limit curves use the exact profiled count objective.

21. 90/120 MeV family: 2016 + 2021 – signed roots and local references



Signed-root curves expose the negative response hidden by positive-signal p_0 . Data use the fixed-mass asymptotic reference; mean-spectrum transforms are diagnostic only. Adjacent masses reuse counts and GP sidebands, so a peak, bridge dip and second peak cannot be assigned independent probabilities and multiplied. No updated calibration of the significance GP or shifted search is claimed.

22. How closely do the fitted means reproduce the data?

The following comparisons use the selected separation and unit strength. Log-limit RMS is calculated at matched available grid coordinates and preserves the response outside the two peaks. The central interval uses the same predeclared central third throughout. The lowest curve discrepancy need not identify the most likely count-spectrum model, because the fitted means and baseline were derived from those same counts.

Fit	Search	Baseline full	Pair full	Pair bridge	Pair outer
L3	2015	0.546	0.545	0.473	0.571
L3	2016	0.539	0.326	0.161	0.356
L3	2021 (10%)	0.332	0.250	0.180	0.285
L3	All three	0.482	0.217	0.189	0.235
L2	2016	0.513	0.303	0.165	0.317
L2	2021 (10%)	0.352	0.257	0.173	0.299
L2	2016 + 2021	0.433	0.223	0.214	0.226
U2	2016	0.620	0.384	0.518	0.287
U2	2021 (10%)	0.303	0.212	0.156	0.227
U2	2016 + 2021	0.429	0.210	0.282	0.233

L3 denotes the lower all-three fit; L2 denotes lower 2016+2021; U2 denotes upper 2016+2021. The lower fits are separate fixed experiments and baseline reconstructions, so their raw objective values must not be compared as if they used the same bins or GP constraint.

For the upper family in 2016, the shared pair's absolute bridge minimum is 2.97×10^{-6} , compared with the observed 1.55×10^{-6} . Its independently refitted alternative reaches 1.69×10^{-6} , but moves the 2021 bridge minimum farther from the observed minimum and fits zero upper-component strength there. Its whole-window 2021 discrepancy nevertheless improves. The upper shared pair improves whole-window shape agreement in both campaigns; reproducing some tightening still leaves a substantial 2016 depth mismatch.

In the lower all-three combination, replacing the postfit continuum by the external-GP continuum worsens the full-window log-limit RMS from 0.217 to 0.610. Conversely, releasing common rates improves the count-fit likelihood by 19.98 but worsens the curve RMS to 0.277. This directly illustrates why a successful limit-curve reconstruction is conditional on its continuum and why that distance is not a signal likelihood.

Several minimum ratios occur at the edge of the central-third interval, especially in 2021 and combined curves. They quantify suppression in that interval; they do not locate a stationary interior trough. Common spacing can organize these checks, but does not establish shared particle production when rates and baseline remain unsettled.

An additive reconstruction offset translates both peaks and leaves their spacing unchanged. A mass-scale effect would also rescale the spacing, and would need a different response model with independent constraints. Beam energy can change production spectra, acceptance and reconstruction bias; it does not by itself shift a particle's invariant mass. Calibration or production explanations therefore require independent detector and event-level controls, beyond these histogram-level fits.

23. Numerical checks, provenance and interpretation

This continuation contains 81,672 recorded upper-limit solutions, 108 completed scenario scans, 18 family mass/rate fits and 14 full-support fitted spectra. No random draws were made. All recorded limits are positive and finite; the maximum absolute CL_s root residual is 2.0×10^{-8} . The maximum profile score is 2.0×10^{-7} . The family validation checks identical fixed experiments, nested mappings, fresh likelihood replay, local mass probes, full-support closure and positivity. The inherited scalar solver passes independent analytic one-bin limits, both asymptotic branches, zero-count cases, derivative checks, covariance-factor rotation and amplitude-unit invariance, and native-data likelihood regressions. Independent reviewers check aligned joint fits, full-support yields, frozen-training identities and numerical replay. These are numerical validations of a conditional model, not coverage or discovery calibrations.

Covariance loading follows the archived factorization: the smallest successful relative addition from 10^{-10} through 10^{-5} times $\max(\max \text{diag } C, 1)$ is used before factorization. Eigenmodes are retained above 10^{-8} in covariance whitened by the Poisson scale. Across baseline contexts, the added diagonal is at most 3.52×10^{-6} of local Poisson variance; representative retrained contexts give a similar maximum. Frozen-training rows report the baseline loading actually used. A metadata-only repair of 1076 diagnostic cells changed no limit or likelihood.

The PDF is rendered page by page and checked for clipped text, and a relocated copy is rebuilt with selected shifted fits replayed from pinned inputs. The source/data archive includes all figures, generated tables, original exported histograms, family spectra, scripts and checksum manifest. The historical ancestry ledger is preserved; the current manifest identifies the files needed by this standalone package.

The earlier 2016 support/qualification exception remains inherited. Resolution, density-to-yield normalization and kernel policies are held fixed. The expanded 2015 scan has not acquired new support calibration. Post-observation region selection, continuum reuse, fitted offsets and multiple model choices remain outside any claimed probability calibration. This continuation does not add toys to the v5.2.1 pilot of ten trials per scheme and strength.

References and source records

- [1] HPS GPR v5.2.1, *Fine mass scans, injected components, and shared peak separations*, 10 September 2026; pinned as `inputs/parent/report.pdf`. This contains the v5.2.0 Section 21 follow-up and the prior injection pilot.
- [2] G. Cowan, K. Cranmer, E. Gross and O. Vitells, *Asymptotic formulae for likelihood-based tests of new physics*, EPJC 71 (2011) 1554, [arXiv:1007.1727](https://arxiv.org/abs/1007.1727).
- [3] M. Frate et al., *Modeling Smooth Backgrounds and Generic Localized Signals with Gaussian Processes*, [arXiv:1709.05681](https://arxiv.org/abs/1709.05681). This supports the general GP modeling approach, not calibration of this selected HPS morphology study.
- [4] Analysis design: `PROTOCOL.md`. Complete mass-fit and response tables accompany the source package in `derived`; independent reviews and numerical checks are in `qa`. The package README identifies each data product and its reproduction command.